

389. Find the difference:

You are given two strings `s` and `t`.

String `t` is generated by random shuffling string `s` and then add one more letter at a random position.

Return the letter that was added to `t`.

Example 1:

Input: `s = "abcd", t = "abcde"`

Output: `"e"`

Explanation: `'e'` is the letter that was added.

Example 2:

Input: `s = "", t = "y"`

Output: `"y"`

Constraints:

- `0 <= s.length <= 1000`
- `t.length == s.length + 1`
- `s` and `t` consists of lowercase English letters.

Solution: time = $O(n)$ and space = $O(1)$

```
class Solution:
    def findTheDifference(self, s: str, t: str) -> str:
        sum_s, sum_t = 0, 0
        for c in s:
            sum_s += ord(c)
        for c in t:
            sum_t += ord(c)
        return chr(sum_t - sum_s)
```

solution: time = $O(n)$ and space = $O(1)$

```
class Solution:
    def findTheDifference(self, s: str, t: str) -> str:
        res = 0

        for c in s:
            res = res ^ ord(c)
        for c in t:
            res = res ^ ord(c)

        return chr(res)
```

solution: more readable and clean with same time and space

```
class Solution:
    def findTheDifference(self, s: str, t: str) -> str:
        res = 0
        for c in s + t:
            res ^= ord(c)
        return chr(res)
```